M-axis Grouped GEMM

One FP8 GEMM over every expert's ragged token batch

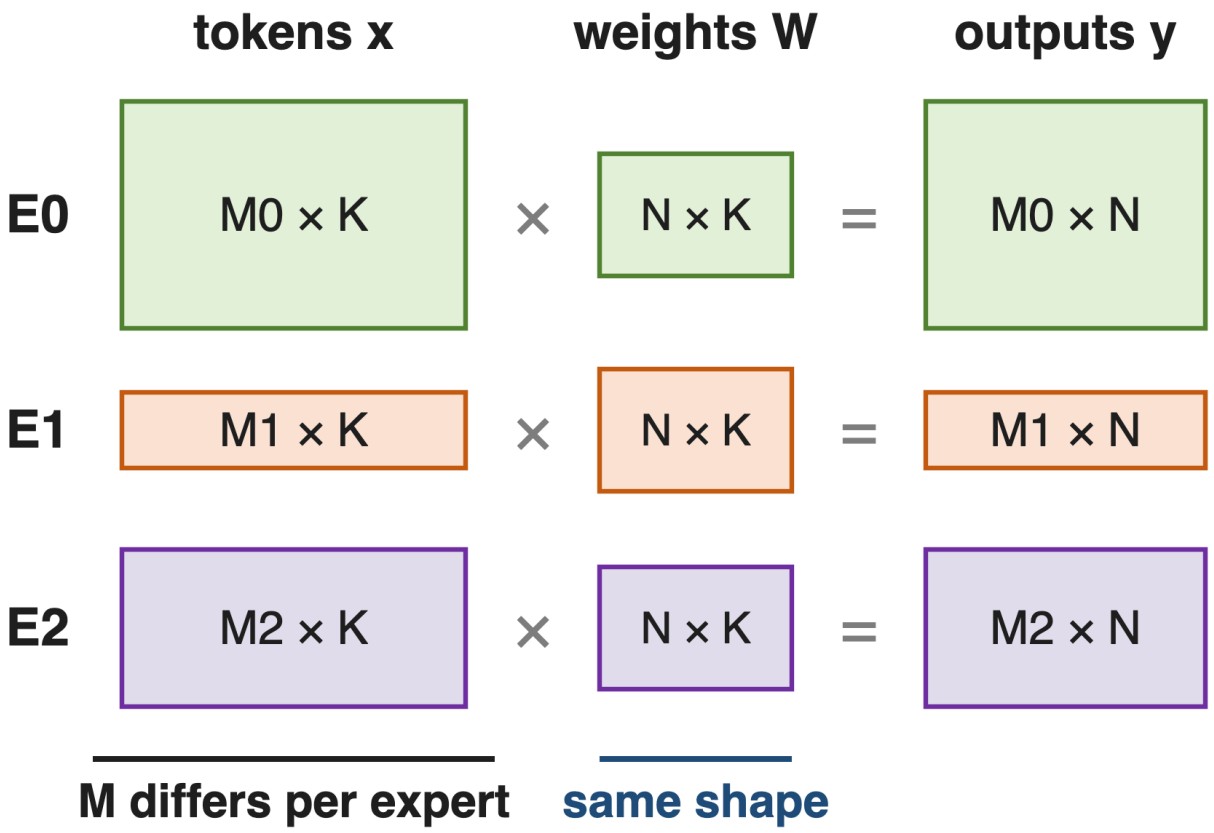
DeepSeek-AI · DeepGEMM (contiguous layout)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

GEMM & MoE Kernels · github.com/deepseek-ai/DeepGEMM

Every MoE layer runs one GEMM per expert

- DeepSeek-V3: **256** routed experts, top-**8**, **58** MoE layers
- Prefill on EP32: **8 + 1** experts per GPU
- Same weight shape, different token count



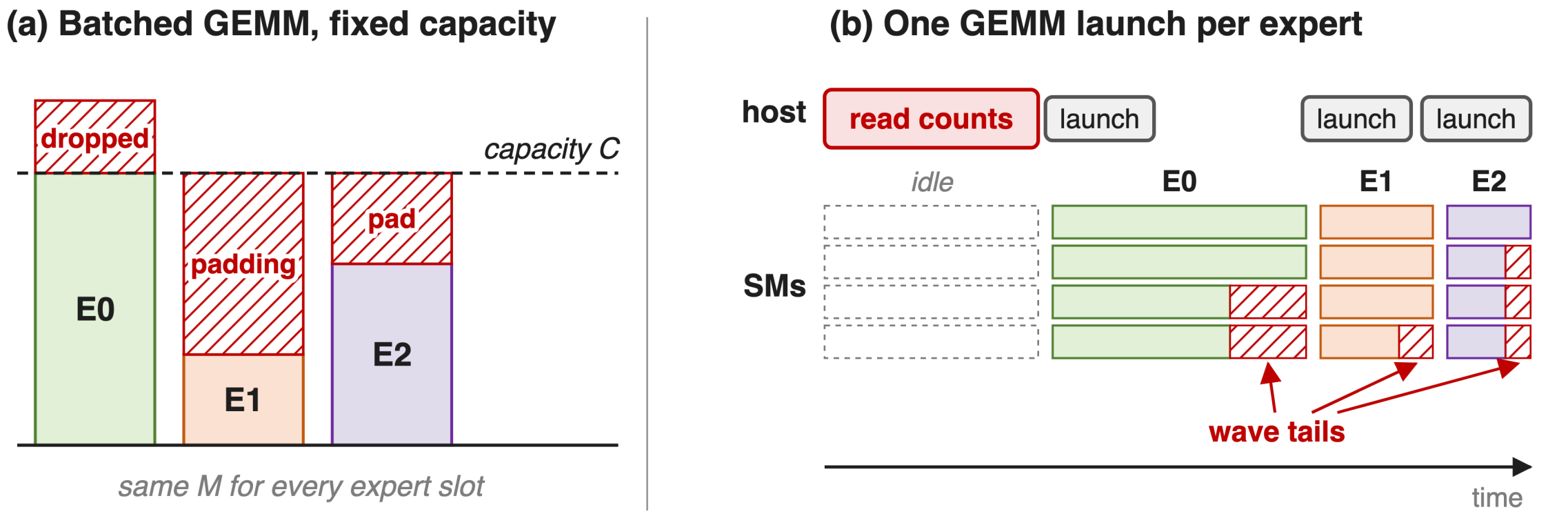
Fixed-shape expert GEMMs pay over 2× the math

> 2×

MoE-layer math needed to stop dropping tokens when every expert gets a fixed-capacity batch

MegaBlocks (MLSys 2023), arXiv 2211.15841, §3 p.4; the same section cites capacity factors as high as 11

Fixed shapes pad, drop, or launch per expert



Capacity batches waste math or tokens; per-expert launches waste SMs and need counts on the host

Insight: experts differ only in M

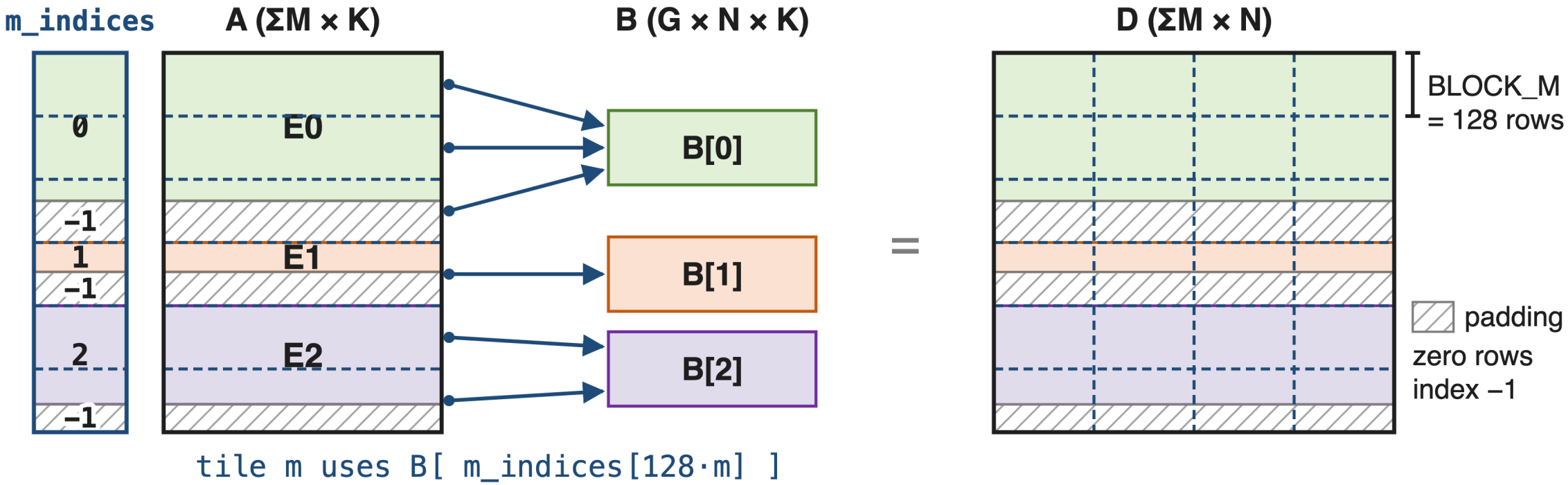
- Every expert: same weight shape `[N, K]`, only `M_e` varies
- Generic grouped GEMMs carry per-problem `(M, N, K)`: unused generality

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- Every expert: same weight shape `[N, K]`, only `M_e` varies
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Stack all tokens along M; if no M tile straddles two experts, one GEMM serves every expert

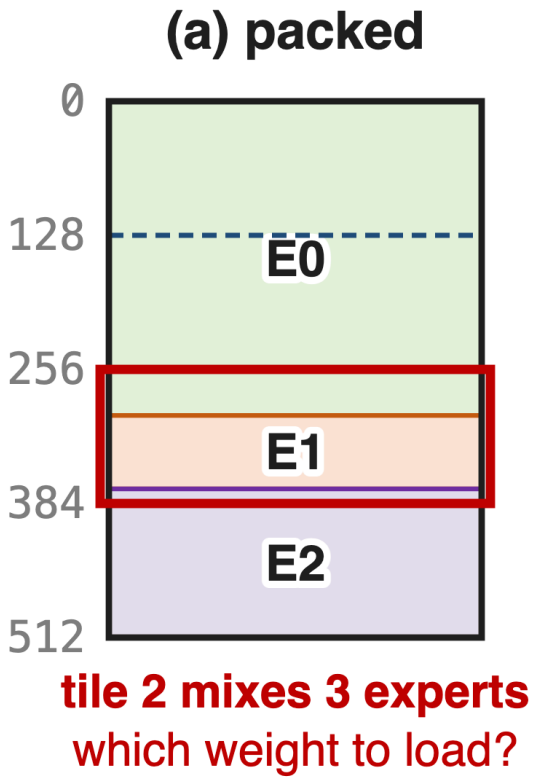
One contiguous layout serves all experts



m_grouped_fp8_gemm_nt_contiguous((A, sfa), (B, sfb), D, m_indices) · A FP8 with 1×128 scales, B FP8 with 128×128 scales, D BF16

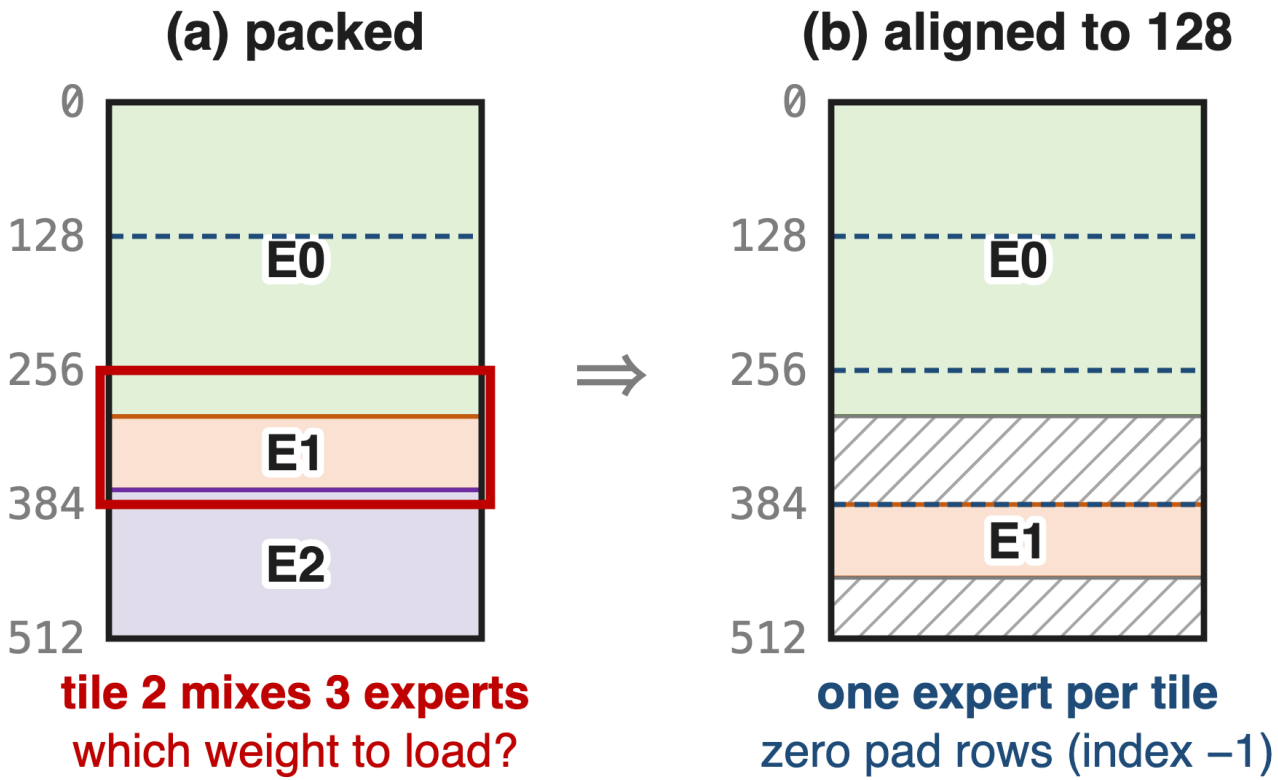
❶ Align every expert segment to the M tile

- An M tile can load only one weight
- Packed rows put **3** experts in tile 2

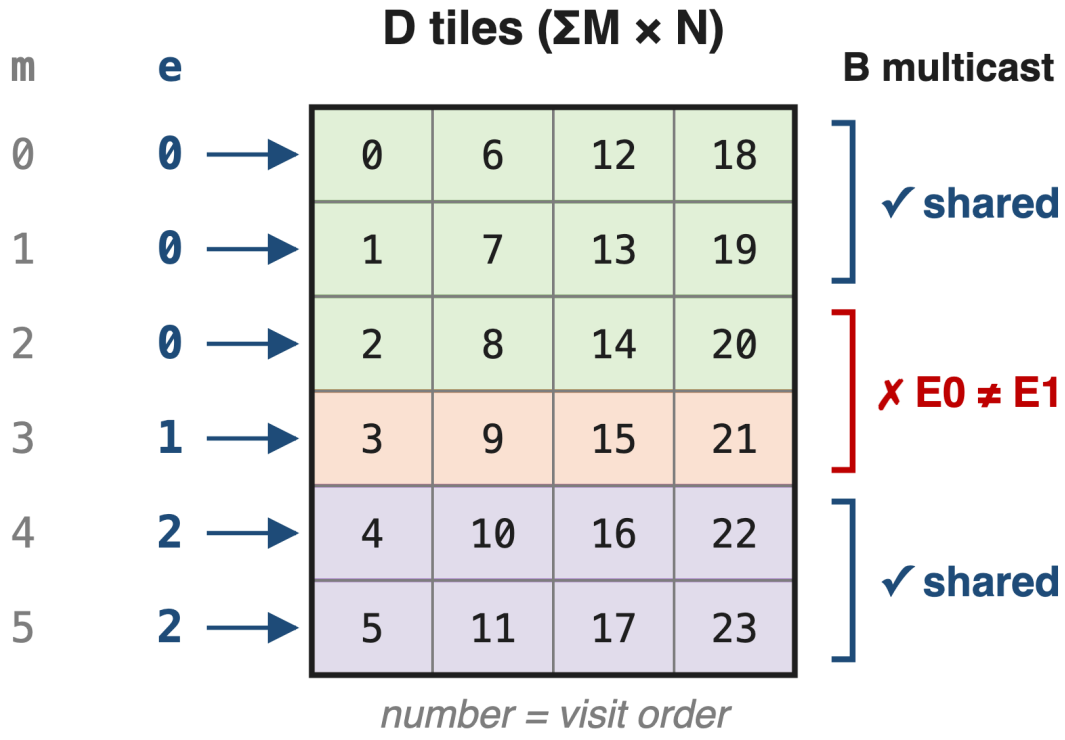


❶ Align every expert segment to the M tile

- An M tile can load only one weight
- Packed rows put **3** experts in tile 2
- Round each expert up to **128** rows (**256** on SM100)
- Pad rows: zero data, index **–1**



❷ Each M tile looks up its own expert



one CTA per SM (persistent)

```
for t in range(sm, tiles, SMs):        all experts
  m, n = swizzle(t)                    down M first
  e = m_indices[m * 128]               expert lookup
  if e < 0: continue                   pure padding
  acc = A[m] @ B[e][n].T               FP8 MMA
  D[m, n] = bf16(acc)                  TMA store
```

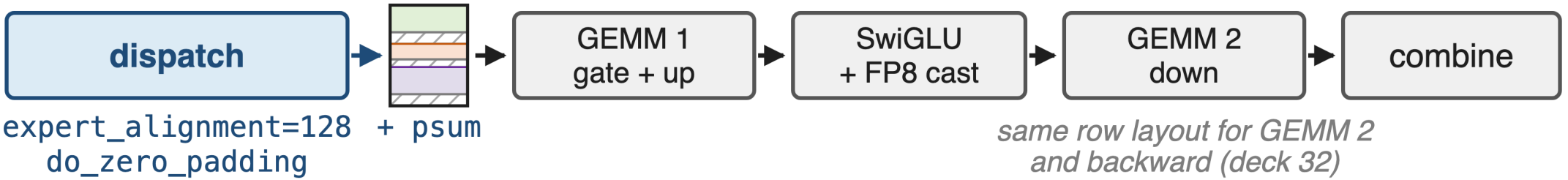
One launch and one tile grid: the host needs only ΣM, never per-expert sizes

❸ Dispatch writes the GEMM-ready layout

(a) DeepEP V1 normal dispatch + permute kernel



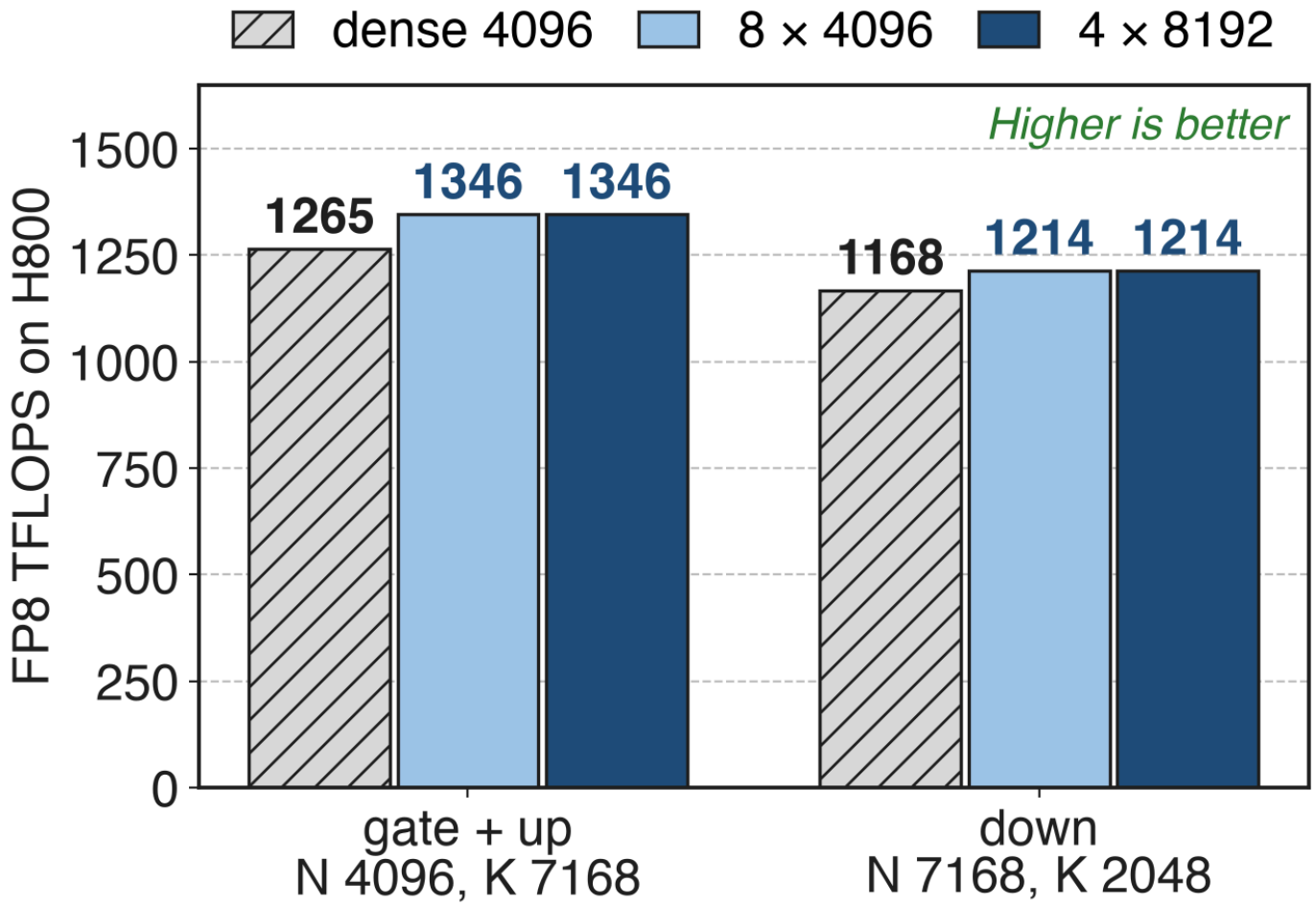
(b) DeepEP V2 dispatch writes the GEMM layout



One aligned buffer feeds gate+up, down, and the backward pass with no extra copy

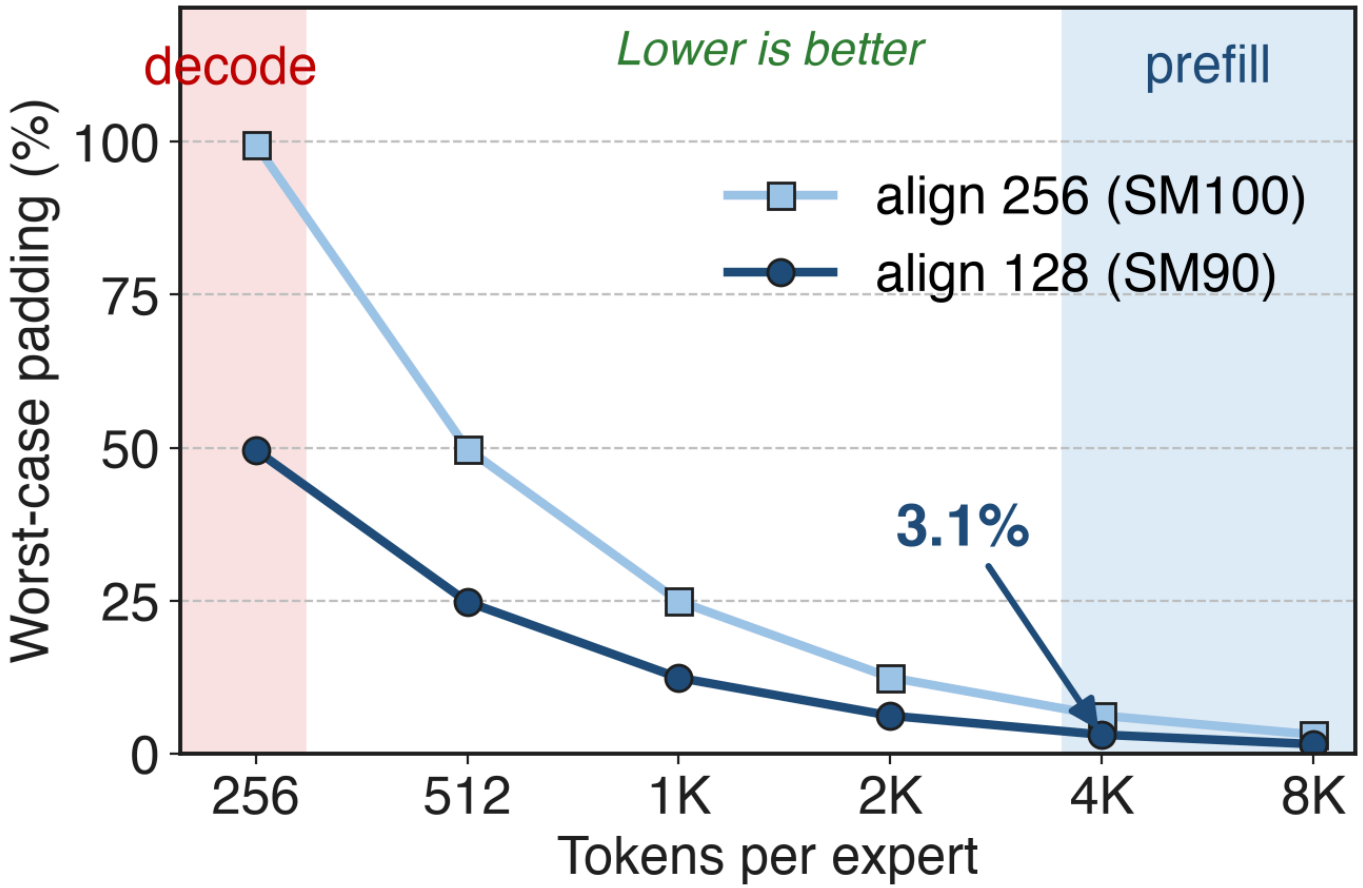
Grouped GEMM matches dense: 1346 TFLOPS

- **1214–1346** TFLOPS on H800
- **1.2–1.3×** over a tuned CUTLASS 3.6 kernel
- Legend: experts × rows per expert



Alignment padding stays ≤ 3.1% at prefill scale

- Worst case **127** pad rows per expert (**255** on SM100)
- **≤ 3.1%** at 4K tokens per expert
- Capacity padding: **> 2×**



Analytic: (alignment − 1) / tokens per expert. Decode: ≤ 256 tokens per expert (DeepSeek-V3 §3.4.2)

Contiguous needs host-side counts and one shape

- **ΣM is a host shape**: counts go to the CPU
- No CUDA Graph: decode uses masked (deck 34)
- One N, K for all experts
- Weight grads group along K (deck 32)
- SM90 FP8: K-major only

FP8×FP4 EP inference: fused Mega MoE replaces this pipeline (deck 35)

Summary

- Fixed-capacity expert GEMMs: **> 2×** math or dropped tokens
- **Insight**: experts differ only in M, so stack rows aligned to **128**
- **1214–1346** TFLOPS on H800, **1.2–1.3×** over CUTLASS
- Padding ≤ one tile per expert: **≤ 3.1%** at 4K tokens

https://github.com/deepseek-ai/DeepGEMM